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FOR

MECHANICAL ENGINEERS

A CERTIFICATE COURSE CONDUCTED
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THE SURE TRUST

Skill Upgradation for Rural-youth Empowerment – Trust

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Major Project:

“WE DESIGNED A RAIL-GUIDED RACK-CLIMBING ROBOTIC SYSTEM FOR
AUTOMATED BIN RETRIEVAL.”

Course Training Attended

By

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NOVEMBER 2025 – APRIL 2026



DECLARATION

This is to certify that Mr. **GAGAN G, NAGESH, RACHANA, POOJA** has successfully completed the four months training given in “**AutoCAD and Solid Works for Mechanical Engineering**” by SURE Trust during the period from November 2025 to February 2026.

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ABSTRACT

This project presents the design and development of a rail-guided rack-climbing robotic system for automated bin retrieval in warehouse environments. With the growing demand for efficient storage and fast order fulfillment, conventional material handling methods face challenges such as limited space utilization, high labour dependency, and reduced operational efficiency. To overcome these issues, the proposed system introduces an autonomous robot capable of moving both horizontally and vertically along storage racks using a guided rail mechanism.

The robot is designed to climb racks and retrieve bins from multiple levels, enabling effective utilization of vertical storage space. Key design considerations include motor selection, torque requirements, load handling capacity, and speed optimization to ensure smooth and reliable operation. The system integrates a compact mechanical structure with a belt or gear-driven climbing mechanism, ensuring stability and precision during movement.

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CHAPTER 1

INTRODUCTION

In recent years, the rapid growth of e-commerce, logistics, and manufacturing industries has significantly increased the demand for efficient warehouse management systems. Traditional storage and material handling methods rely heavily on manual labour, which leads to higher operational costs, lower efficiency, and increased chances of human error. As a result, industries are shifting towards automation to improve productivity, accuracy, and speed in warehouse operations.

Automated Storage and Retrieval Systems (ASRS) and Autonomous Mobile Robots (AMRs) have been widely adopted to address these challenges. However, conventional systems often face limitations such as high installation costs, lack of flexibility, and inefficient utilization of vertical storage space. This creates a need for innovative solutions that can overcome these drawbacks.

1.1 Rack Mechanism

A clean engineering style 3D render of a rack and pinion mechanism used in a warehouse climbing robot, isolated on white background. Show two round metal pinion gears engaging with a long straight rack gear. Include mounting frame and motor housing. Add clear labels with arrows: "Pinion Gear 1", "Pinion Gear 2", "Rack Gear", "Motor Mount", "Bearing Support". High detail, realistic, technical drawing style.

1.2 Scissor Lift Mechanism

A detailed 3D engineering illustration of a scissor lift mechanism used in a robotic system, isolated on white background. Show X-shaped metal arms, pivot joints, base frame, and top platform with a linear actuator. Add labeled arrows: "Scissor Arms", "Pivot Joint", "Linear Actuator", "Base Frame", "Top Platform". Clean technical diagram style.

1.3 Robot Base

A 3D render of a robotic mobile base chassis for an automated warehouse robot, isolated on white background. Show wheels, motors, battery compartment, and frame. Add labels: "Drive Wheel", "Motor", "Battery Pack", "Chassis Frame". Engineering presentation style, high clarity.

1.4 Bin Handling Unit

A technical 3D illustration of a bin handling mechanism for a warehouse robot, isolated on white background. Show sliding rails and a platform used to hold bins. Add labels: "Bin Holder", "Sliding Rail", "Actuator". Clean and minimal engineering diagram.

1.5 PCB & Control Unit

A detailed 3D render of a robot control unit with exposed PCB board inside a protective casing, isolated on white background. Show wiring ports and electronics. Add labels: "PCB Board", "Controller Unit", "Power Module", "Connectors". Professional technical illustration.

1.6 Sensor Module

A close-up 3D render of robot sensors used in automation, isolated on white background. Include proximity sensors, camera module, encoder, and limit switches. Add labels: "Proximity Sensor", "Vision Camera", "Encoder", "Limit Switch". High-detail engineering style.

1.7 Outer Body Covers

A 3D illustration of robot outer body panels and covers, isolated on white background. Show top cover, side panels, and screws. Add labels: "Top Cover", "Side Panel", "Mounting Screws". Clean technical drawing style.

1.8 Exploded View

An exploded view of a warehouse climbing robot showing all components separated in layers, including rack mechanism, scissor lift, base, PCB, sensors, and covers. Each part clearly spaced and labeled with arrows. White background, professional engineering diagram.

CHAPTER 2

LITERATURE SURVEY

2.1 Hai Pick Climb System (Hai Robotics)

The Hai Pick Climb system is an advanced Automated Storage and Retrieval System (ASRS) that uses climbing robots for goods-to-person operations. The system enables robots to move both vertically and horizontally using a single-sided guide rail, reducing installation complexity and improving efficiency. It achieves high throughput of up to 4000 totes per hour within 1000 m² and supports vertical storage up to 12 meters.

Information Taken from this Study:

- Concept of rack-climbing robot mechanism
- Performance parameters such as speed, throughput, and accuracy
- System scalability (up to 3000 robots)
- Design idea for vertical movement using guide rails

2.2 Traditional ASRS (Crane-Based Systems)

Traditional ASRS systems use stacker cranes, conveyors, and fixed storage systems for material handling. These systems are widely used in large warehouses but require high precision infrastructure and involve high installation costs. They are less flexible and difficult to expand once installed.

Information Taken from this Study:

- Limitations of conventional automation systems
- Need for flexible and scalable solutions
- Comparison base for evaluating modern robotic systems
- Understanding of structured warehouse automation

2.3 Autonomous Mobile Robots (AMRs)

Autonomous Mobile Robots (AMRs) are used for horizontal transportation of goods in warehouses. They improve efficiency by reducing human labour and increasing speed. However, AMRs are limited to ground-level movement and cannot effectively utilize vertical storage space.

Information Taken from this Study:

- Working principle of mobile robots
- Limitation in vertical movement
- Importance of combining vertical and horizontal mobility
- Base concept for robot navigation and automation

2.4 Rack-Climbing Robotic Systems in Warehouses

Recent developments in warehouse automation include rack-climbing robots that can access inventory at different heights. These systems use rails or guided mechanisms to climb storage racks, allowing efficient use of vertical space. They also reduce travel time and improve picking efficiency.

Information Taken from this Study:

- Design concept of climbing mechanism
- Use of guide rails for stability
- Improved space utilization techniques
- Inspiration for developing climbing robot system

CHAPTER 3

Problem Statement

Existing warehouse automation systems are not fully efficient in utilizing available space, especially vertical storage. Many systems are restricted to horizontal movement or require complex infrastructure for vertical access. Additionally, manual intervention in bin retrieval leads to delays, errors, and increased labour costs. Therefore, there is a need to develop a system that can efficiently access storage at different heights while maintaining high speed, accuracy, and reliability.

CHAPTER 4

Objective of the Project

The main objective of this project is to design and develop a rail-guided rack-climbing robotic system for automated bin retrieval.

The specific objectives include:

- To design a robot capable of both horizontal and vertical movement
- To develop a rail-guided climbing mechanism for rack navigation
- To improve warehouse space utilization by accessing vertical storage
- To reduce human effort and increase operational efficiency
- To achieve accurate and fast bin retrieval

CHAPTER 5

Scope of the Project

This project focuses on the conceptual design and analysis of a rack-climbing robotic system used in warehouse automation. It includes the study of mechanical design, motor selection, torque calculation, and system performance. The system can be applied in industries such as e-commerce warehouses, logistics Centers, manufacturing units, and distribution hubs. The proposed system is scalable and can be extended for large warehouse operations by integrating multiple robots and advanced control systems.

CHAPTER 6

***Technical Performance Benchmarks**

Metric	Unit	Performance Value
Max Ground Travel Speed	m/s	4.0
Max Vertical Climbing Speed	m/s	1.0
System Capacity	Robots	Up to 3,000
Throughput Density	Totes/Hr/1k m ²	4,000
Vertical Reach	Meters	12.0
Pick Accuracy	Percentage	99.9%

CHAPTER 7

Calculations

Total mass = 60 kg,

gravity = 9.81 m/s^2 ,

wheel radius = 0.05 m,

efficiency = 80%.

Force required for climbing: $F = m \times g = 60 \times 9.81 = 588.6 \text{ N}$.

Torque required: $T = F \times r = 588.6 \times 0.05 = 29.43 \text{ Nm}$.

Power required: $P = F \times v = 588.6 \times 1 = 588.6 \text{ W}$.

Considering efficiency: Actual power $\approx 750 \text{ W}$.

Motor speed $\approx 200 \text{ RPM}$,

Gear ratio $\approx 15:1$

Parameter	Value
Force	589 N
Torque	30 Nm
Power	750 W
Speed	200 RPM
Gear Ratio	15:1

CHAPTER 8

Working Principle

1. Robot moves to rack location
2. Climbs using rack mechanism
3. Positions using sensors
4. Lifts using scissor mechanism
5. Retrieves bin
6. Returns to base

Advantages

- High efficiency
- Reduced labour
- Space saving
- High accuracy

Applications

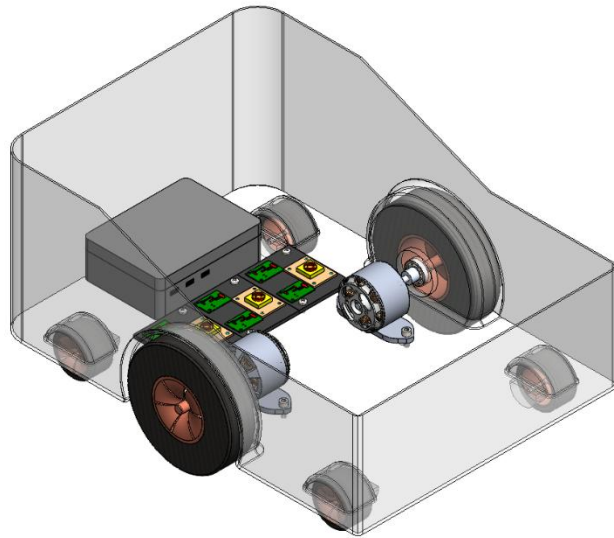
- Warehouses
- E-commerce logistics
- Manufacturing storage

Future Scope

- AI integration
- Autonomous navigation
- Improved sensors

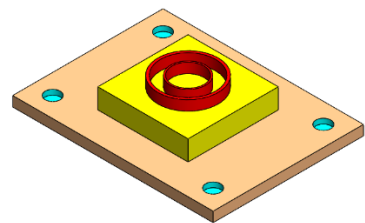
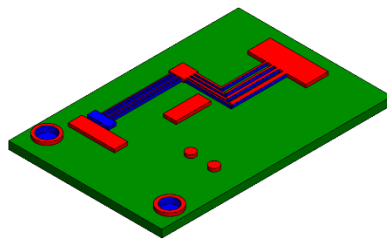
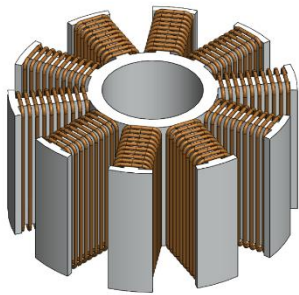
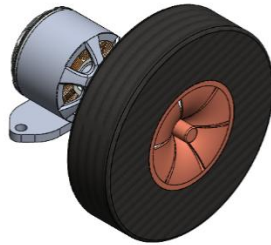
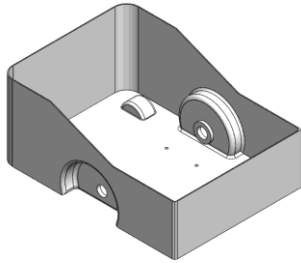
CHAPTER 9

9.1 Subassembly of Lower Body & Part



The lower body sub-assembly acts as the base of the robot and supports all the important components. It includes the frame, wheels, motor, battery, PCB, and support rollers. The motor helps in rotating the wheels, which allows the robot to move smoothly. The support rollers maintain balance and reduce friction during movement. The PCB controls the overall operation and manages the power supply from the battery. Overall, this assembly provides stability, proper support, and smooth movement of the robot

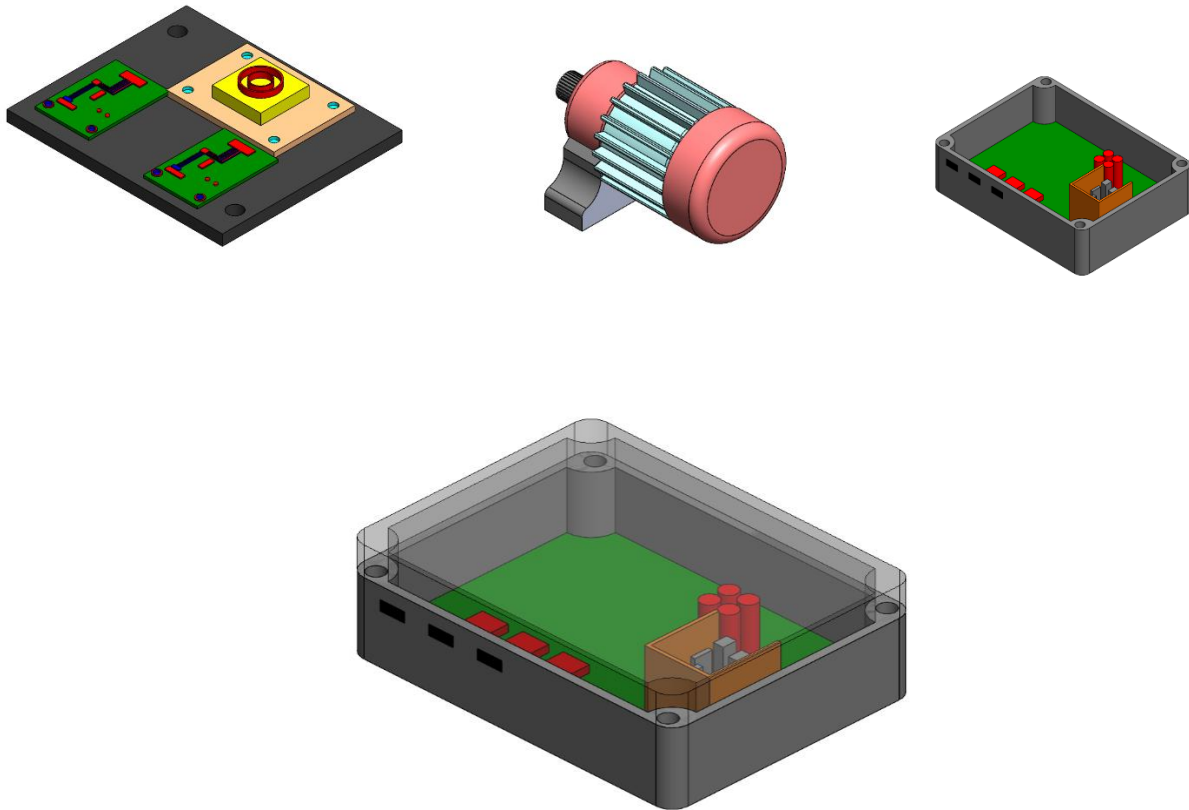
PARTS:-



The lower body consists of several important parts that work together to support and move the robot. It includes the base frame which holds all the components in position. The motor and wheels are responsible for the movement of the robot. The rollers help in maintaining balance and smooth motion. The coil (winding part) is used in the motor for generating motion. The PCB controls the electrical connections and overall functioning of the system. A mounting plate is also used to fix and align components properly.

All these parts are assembled together to provide strength, stability, and efficient movement to the robot.

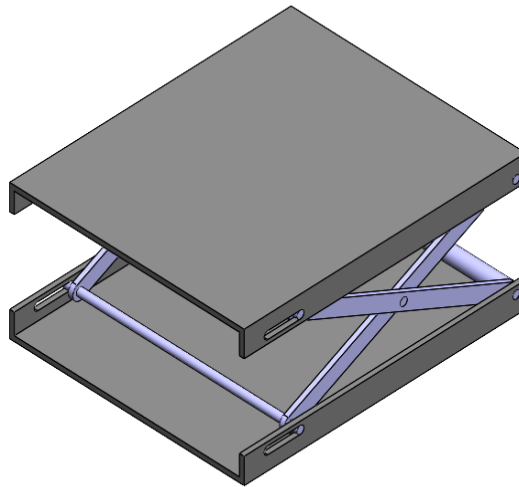
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This assembly includes the PCB, Arduino, motor, and enclosure box, which together control the robot's operation. The Arduino acts as the main controller and sends signals to different components. The PCB is used to manage electrical connections and ensure proper power distribution. The motor converts electrical energy into mechanical motion to drive the robot. All these components are placed inside a protective enclosure, which keeps them safe and properly arranged.

Overall, this assembly controls the movement and functioning of the robot in a smooth and efficient way.

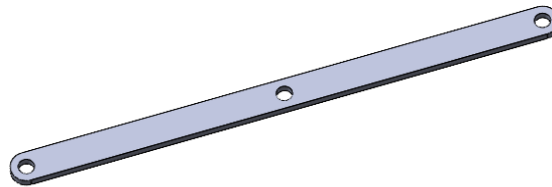
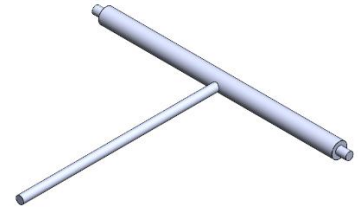
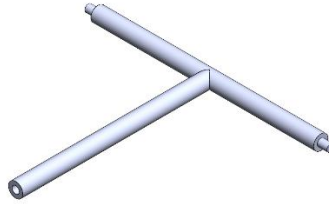
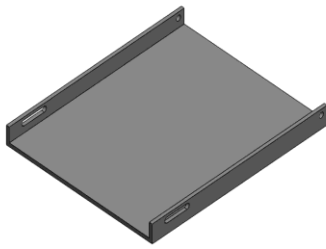
9.2 CROSS 'X' Assembly



The scissor lift sub-assembly is used to adjust the height of the robot's upper platform. It consists of a base plate, a top plate, and crossed link arms connected by pivot joints, forming a scissor mechanism. When force is applied (by a motor or actuator), the links expand or contract, causing the top platform to move up or down.

This mechanism provides stable and controlled vertical movement while maintaining compact size when closed. It is usually made of mild steel or aluminum for strength and durability. Overall, this sub-assembly helps in smooth height adjustment and proper support of mounted components.

PARTS:-



Base & Top Plate

The base and top plate form the main support of the scissor lift. The base is fixed to the robot body, and the top plate carries the load. They provide stability and proper alignment.

Tube Structure

The tube structure forms the crossed arms of the scissor mechanism. It allows smooth lifting and lowering while maintaining strength with less weight.

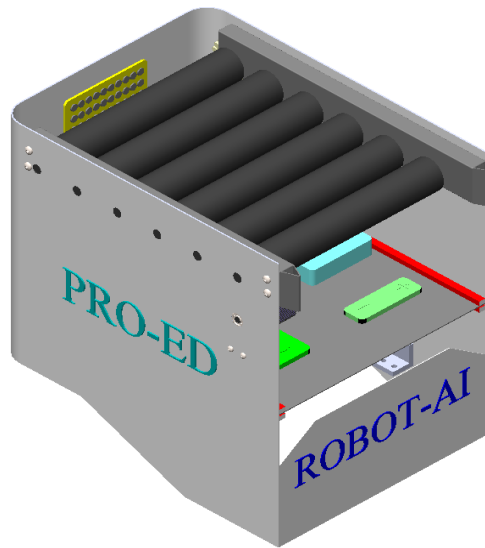
Rod Structure

The rod connects the pivot points and helps transfer motion evenly. It ensures proper alignment and stable vertical movement.

Support Strip

The support strip strengthens the mechanism and prevents bending. It improves stability and durability during operation.

9.3 SUB ASSEMBLY OF UPEER BODY

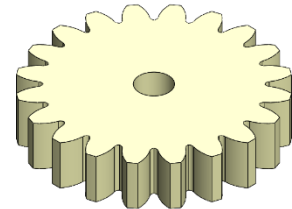
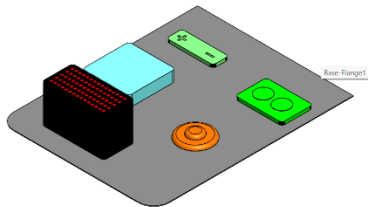
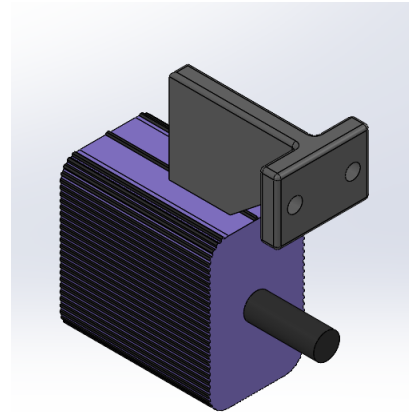
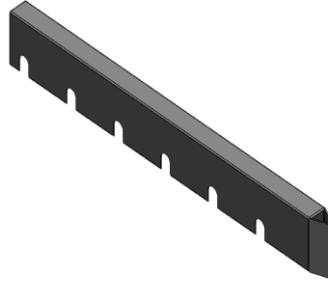
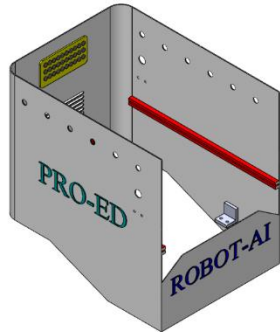


Upper Body Sub-Assembly

The upper body sub-assembly forms the top structure of the robot and houses important components such as rollers, electronic modules, and supporting frames. It provides protection to internal parts and maintains proper alignment of the climbing mechanism.

The rollers mounted on the upper section help in smooth movement along the rail and improve stability during operation. The rigid outer casing ensures strength, safety, and durability. Overall, this sub-assembly provides structural support, component protection, and stable climbing performance.

PARTS:-



***Upper Body**

The upper body is the main outer structure that protects internal components and supports the roller mechanism.

***It protects the rollers and internal parts from dust and damage.**

***Motor**

The motor provides rotational motion to drive the robot for climbing.

***Accessories Plate**

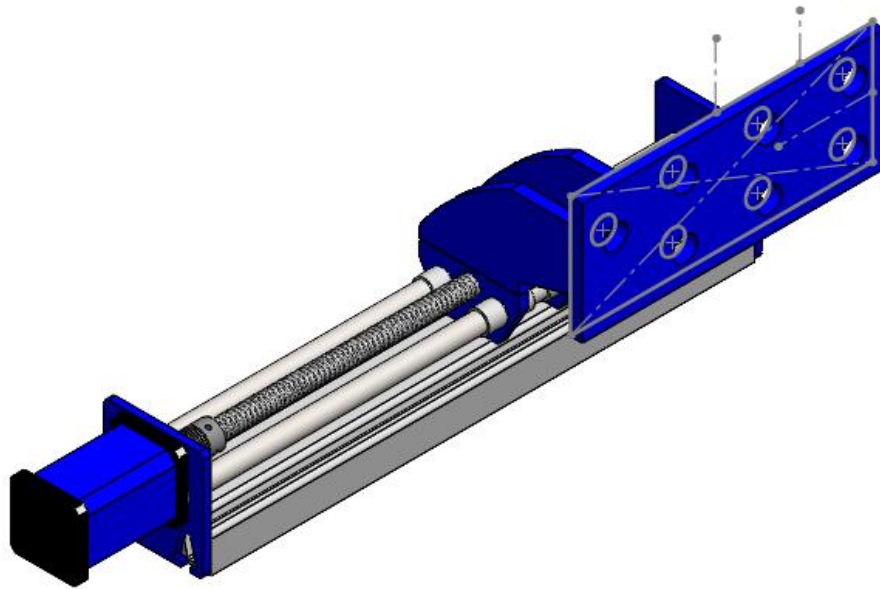
It holds electronic components like battery and PCB in proper position.

***Lower Body**

The lower body acts as the base support and maintains stability.

***Gear:-** The gear transfers motor power and increases torque for smooth climbing.

PARTS:-



***Linear Sliding System Sub-Assembly**

The linear sliding system is used to provide controlled linear motion in the robot. It consists of guide rails, a lead screw, sliding block, and a motor for actuation. When the motor rotates the lead screw, the sliding block moves forward or backward along the guide rails.

This mechanism ensures smooth, accurate, and stable movement of mounted components. It improves positioning control and overall operational efficiency of the robot.

9.4 SUB ASSEMBLY OF ROBOT ARM

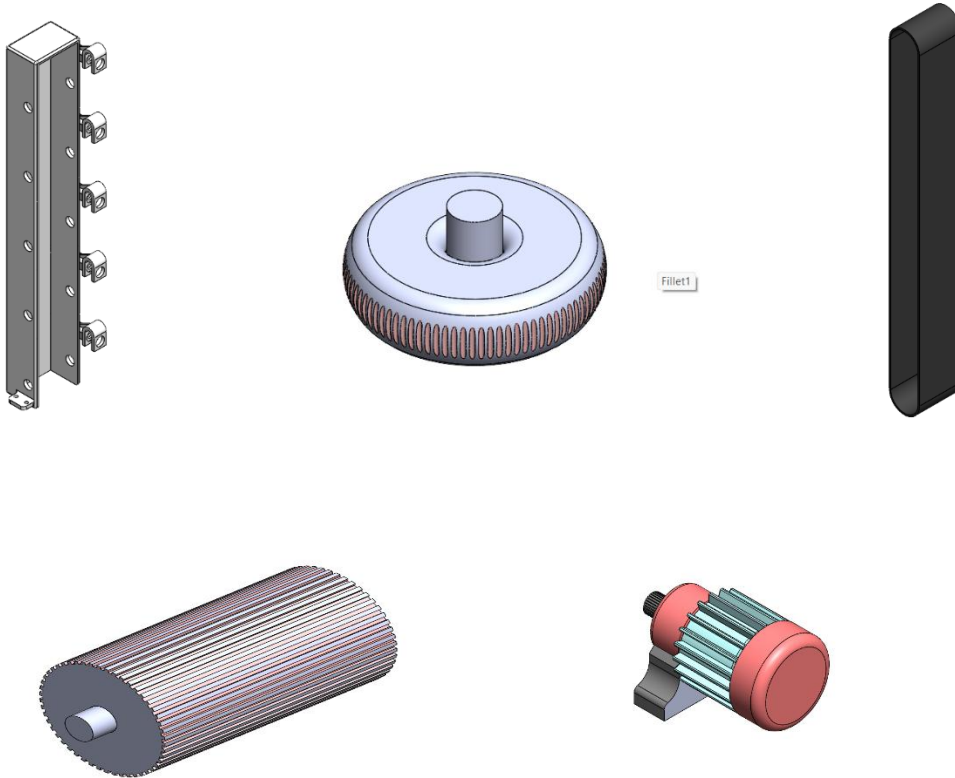


***Robot Arm Sub-Assembly**

The robot arm sub-assembly is used to support and guide the robot along the rail structure. It consists of a vertical frame with multiple guide rollers mounted on one side and a motor at the base for actuation.

The rollers ensure smooth contact with the rail and maintain proper alignment during climbing. The rigid frame provides strength and stability, while the motor enables controlled movement. Overall, this sub-assembly ensures balanced, stable, and efficient rail-guided motion of the robot.

PARTS:-



***Main Frame:-** The main frame is the structural backbone of the robot arm assembly. It supports all components such as rollers, motor, and belt system, and maintains proper alignment and rigidity during operation.

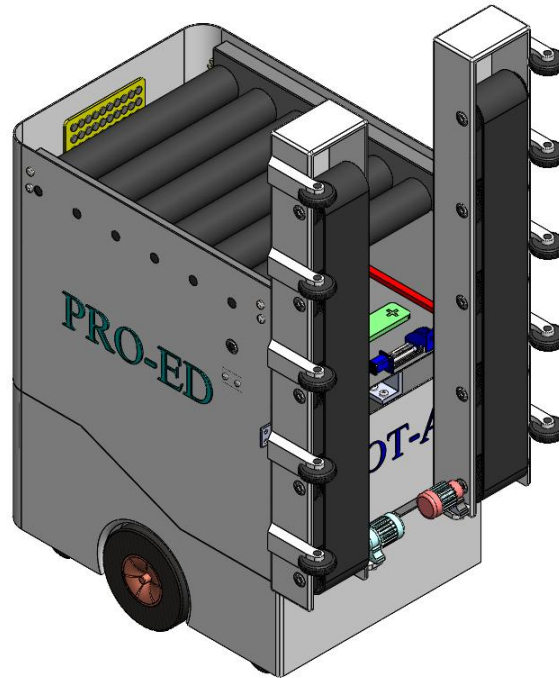
***Mini Roller:-** The mini rollers guide the arm along the rail and reduce friction. They help maintain balance and ensure smooth and stable movement.

***Belt:-** The belt is used to transmit power from the motor to the roller system. It ensures smooth motion and efficient power transfer.

***Roller with Gripper:-** The roller with gripper provides traction against the rail surface. The gripper surface increases friction, allowing the robot to climb securely without slipping.

***Motor:-** The motor drives the roller mechanism through the belt system. It provides the required torque and speed for climbing movement.

9.5 Rail guided climbing robot assembly design



The total assembly consists of the lower body, upper body, robot arm, roller mechanism, motor drive system, and electronic components integrated into a single compact structure. The main frame supports all sub-assemblies and maintains proper alignment during operation.

The motor-driven roller system provides the required traction to climb along the rail, while the guide rollers ensure stability and smooth movement. The belt and gear mechanism transmit power efficiently from the motor to the rollers. Electronic components such as the battery and PCB control the overall operation of the robot.

Overall, the complete assembly ensures stable, controlled, and efficient rail-guided climbing performance.

CHAPTER 10

Conclusion

This project focused on the design and analysis of a rail-guided rack-climbing robotic system for automated bin retrieval in warehouse environments. The study highlighted the limitations of traditional material handling systems and emphasized the need for advanced automation solutions to improve efficiency, accuracy, and space utilization.

The proposed system successfully demonstrates the concept of a robot capable of both horizontal and vertical movement using a guided rail mechanism. By integrating climbing capability with automated retrieval, the system effectively utilizes vertical storage space, which is often underused in conventional warehouses. Key design aspects such as motor selection, torque requirements, load handling, and speed were considered to ensure reliable and efficient operation.

CHAPTER 11

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